

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
```

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```
uploaded = files.upload()
```

 No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving ChatGPT Image Aug 19, 2026, 11:28:10 PM.png to ChatGPT Image Aug 19, 2026, 11:28:10 PM (1).png

```
image = cv2.imread(next(iter(uploaded)))

# Convert BGR to RGB for Matplotlib
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8,6))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Original Image



```
height, width, channels = image.shape

print("Image Properties")
print("-----")
print("Width:", width, "pixels")
print("Height:", height, "pixels")
print("Resolution:", width, "x", height)
print("Number of Channels:", channels)
print("Data Type:", image.dtype)
```

```
Image Properties
-----
Width: 1536 pixels
Height: 1024 pixels
Resolution: 1536 x 1024
Number of Channels: 3
Data Type: uint8
```

```
cv2.imwrite("image_output.jpg", image)
cv2.imwrite("image_output.png", image)

print("Image saved successfully!")
print("JPEG file: image_output.jpg")
print("PNG file: image_output.png")
```

```
Image saved successfully!
JPEG file: image_output.jpg
PNG file: image_output.png
```

```
jpeg_image = cv2.imread("image_output.jpg")
png_image = cv2.imread("image_output.png")

jpeg_image = cv2.cvtColor(jpeg_image, cv2.COLOR_BGR2RGB)
png_image = cv2.cvtColor(png_image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.imshow(jpeg_image)
plt.title("JPEG Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(png_image)
plt.title("PNG Image")
plt.axis("off")

plt.show()
```

JPEG Image



PNG Image



```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(8,6))
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
```

Grayscale Image



```
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)

# Convert HSV to RGB only for proper display
hsv_rgb = cv2.cvtColor(hsv, cv2.COLOR_HSV2RGB)

plt.figure(figsize=(8,6))
plt.imshow(hsv_rgb)
plt.title("HSV Image")
plt.axis("off")
plt.show()
```

HSV Image



```
lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)

# Convert LAB to RGB for display
lab_rgb = cv2.cvtColor(lab, cv2.COLOR_LAB2RGB)

plt.figure(figsize=(8,6))
plt.imshow(lab_rgb)
plt.title("LAB Image")
plt.axis("off")
plt.show()
```

LAB Image



```
plt.figure(figsize=(15,5))

plt.subplot(1,4,1)
plt.imshow(image_rgb)
plt.title("Original")
plt.axis("off")

plt.subplot(1,4,2)
plt.imshow(gray, cmap="gray")
```

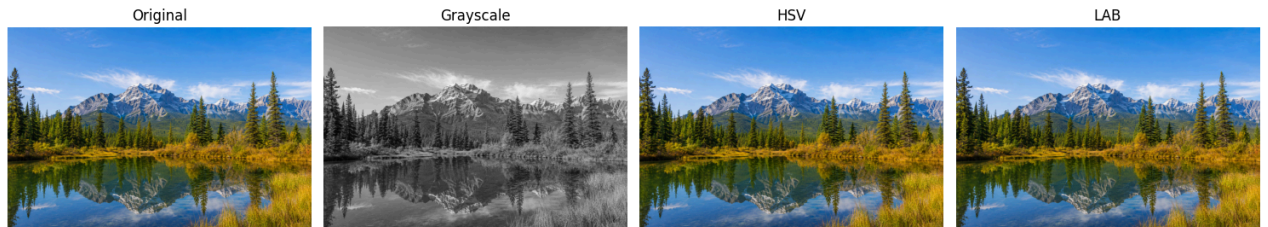


```
plt.title("Grayscale")
plt.axis("off")

plt.subplot(1,4,3)
plt.imshow(hsv_rgb)
plt.title("HSV")
plt.axis("off")

plt.subplot(1,4,4)
plt.imshow(lab_rgb)
plt.title("LAB")
plt.axis("off")

plt.tight_layout()
plt.show()
```



```
resized = cv2.resize(image, (400, 300))

resized_rgb = cv2.cvtColor(resized, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8,6))
plt.imshow(resized_rgb)
plt.title("Resized Image (400 x 300)")
plt.axis("off")
plt.show()
```



```
height, width = image.shape[:2]

center = (width // 2, height // 2)

rotation_matrix = cv2.getRotationMatrix2D(
    center,
    45,
    1.0
)

rotated = cv2.warpAffine(
    image,
    rotation_matrix,
```



```
(width, height)
)

rotated_rgb = cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8,6))
plt.imshow(rotated_rgb)
plt.title("Rotated Image - 45 Degrees")
plt.axis("off")
plt.show()
```

Rotated Image - 45 Degrees



```
horizontal_flip = cv2.flip(image, 1)

horizontal_flip_rgb = cv2.cvtColor(
    horizontal_flip,
    cv2.COLOR_BGR2RGB
)

plt.figure(figsize=(8,6))
plt.imshow(horizontal_flip_rgb)
plt.title("Horizontal Flip")
plt.axis("off")
plt.show()
```

Horizontal Flip



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```
vertical_flip = cv2.flip(image, 0)

vertical_flip_rgb = cv2.cvtColor(
    vertical_flip,
```

```

        cv2.COLOR_BGR2RGB
    )

plt.figure(figsize=(8,6))
plt.imshow(vertical_flip_rgb)
plt.title("Vertical Flip")
plt.axis("off")
plt.show()

```

Vertical Flip



```

plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.imshow(horizontal_flip_rgb)
plt.title("Horizontal Flip")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(vertical_flip_rgb)
plt.title("Vertical Flip")
plt.axis("off")

plt.tight_layout()
plt.show()

```

Horizontal Flip



Vertical Flip



```

negative = 255 - image

negative_rgb = cv2.cvtColor(
    negative,
    cv2.COLOR_BGR2RGB
)

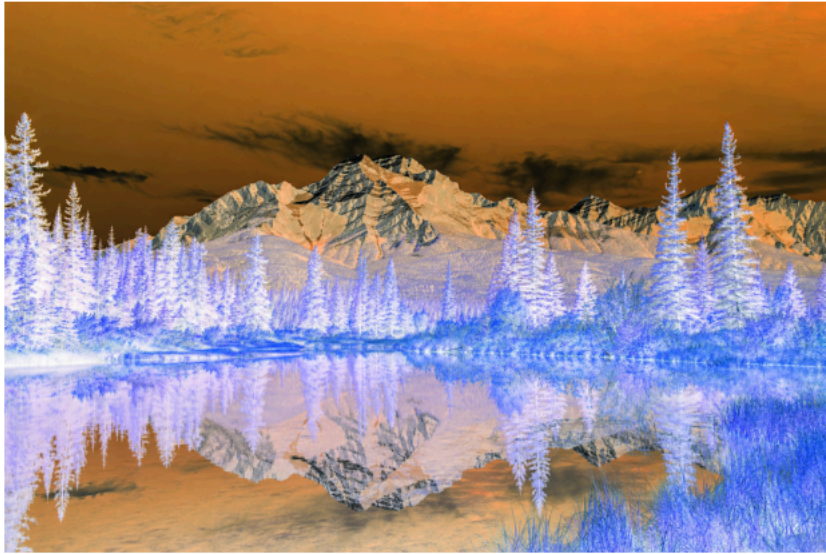
plt.figure(figsize=(8,6))
plt.imshow(negative_rgb)

```



```
plt.title("Negative / Complement Image")
plt.axis("off")
plt.show()
```

Negative / Complement Image



```
height, width = image.shape[:2]

# Select central region
x1 = width // 4
y1 = height // 4
x2 = 3 * width // 4
y2 = 3 * height // 4

roi = image[y1:y2, x1:x2]

roi_rgb = cv2.cvtColor(roi, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8,6))
plt.imshow(roi_rgb)
plt.title("Region of Interest (ROI)")
plt.axis("off")
plt.show()
```

Region of Interest (ROI)



```
plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(roi_rgb)
```



```
plt.title("Region of Interest")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Original Image



Region of Interest



```
plt.figure(figsize=(16,12))

plt.subplot(3,4,1)
plt.imshow(image_rgb)
plt.title("Original")
plt.axis("off")

plt.subplot(3,4,2)
plt.imshow(gray, cmap="gray")
plt.title("Grayscale")
plt.axis("off")

plt.subplot(3,4,3)
plt.imshow(hsv_rgb)
plt.title("HSV")
plt.axis("off")

plt.subplot(3,4,4)
plt.imshow(lab_rgb)
plt.title("LAB")
plt.axis("off")

plt.subplot(3,4,5)
plt.imshow(resized_rgb)
plt.title("Resized")
plt.axis("off")

plt.subplot(3,4,6)
plt.imshow(rotated_rgb)
plt.title("Rotated")
plt.axis("off")

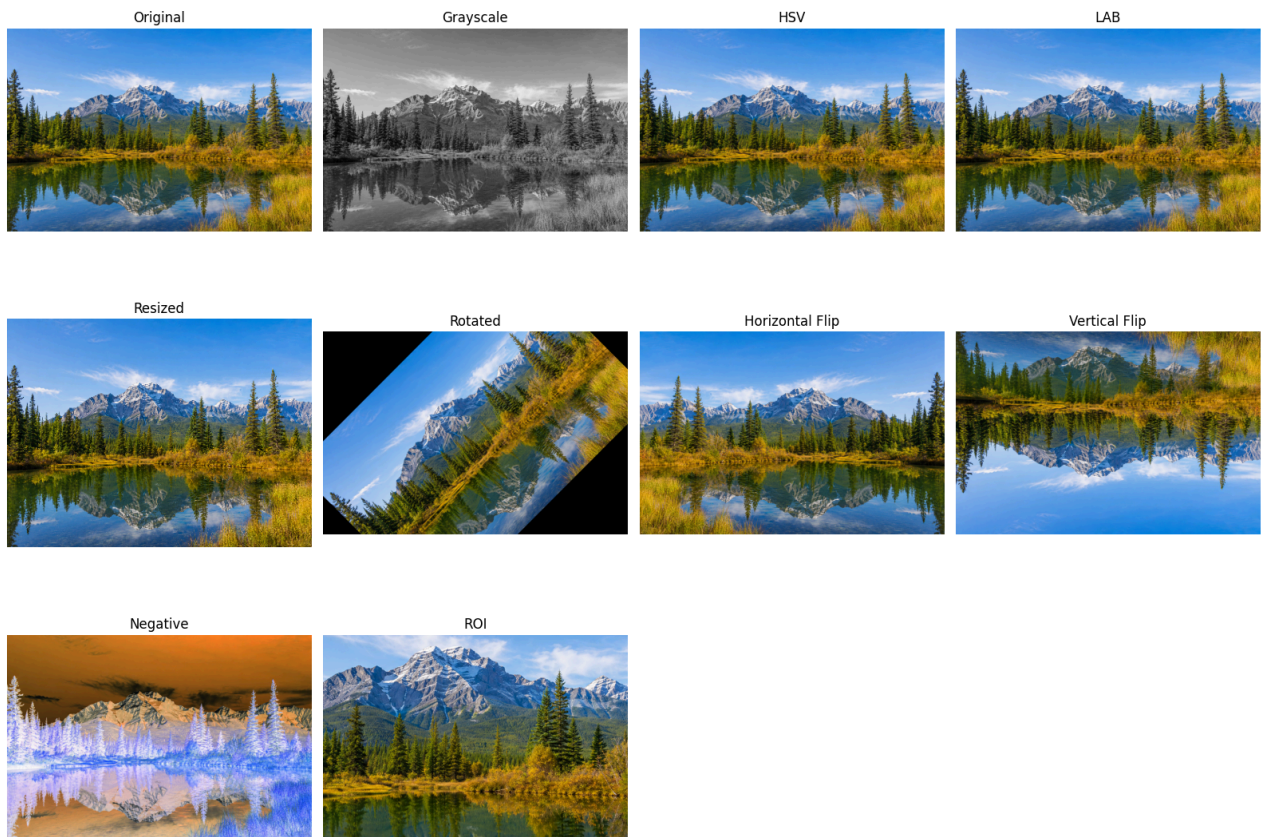
plt.subplot(3,4,7)
plt.imshow(horizontal_flip_rgb)
plt.title("Horizontal Flip")
plt.axis("off")

plt.subplot(3,4,8)
plt.imshow(vertical_flip_rgb)
plt.title("Vertical Flip")
plt.axis("off")

plt.subplot(3,4,9)
plt.imshow(negative_rgb)
plt.title("Negative")
plt.axis("off")

plt.subplot(3,4,10)
plt.imshow(roi_rgb)
plt.title("ROI")
plt.axis("off")
```

```
plt.tight_layout()  
plt.show()
```



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